MAT 110

Lecture-01 (Section 1.1 & 1.2 H.A calculus, 10th edition)

* **Limit:** If the values of $f(x)$ can be made as close as we like to L by taking values of x sufficiently close to a (but not equal to a) then we write

$$\lim_{x \to a} f(x) = L$$

Example-1

let $f(x) = \dfrac{x^2-9}{x-3}$

$$\therefore \lim_{x \to 3} f(x)$$

$$= \lim_{x \to 3} \frac{x^2-9}{x-3}$$

x	$f(x)$	x	$f(x)$
2.99	5.99	3.01	6.01
2.999	5.999	3.001	6.001
2.9999	5.9999	3.00001	6.00001
⋮			
2.999999	5.999999	3.000001	6.000001

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~~But~~ here, we observe that the func $f(x)$ is undefined at $x=3$. But, when we move closer to 3 the value of the function moves toward 6. That means the limiting values of the function is 6 when the value of x moves toward 3.

★ Example-02

$$\lim_{x \to 2} \frac{5x^3+4}{x-3}$$

$$= \frac{5 \cdot 2^3+4}{2-3} = -44$$

★ L.H.L (Left Hand limit)

If the values of $f(x)$ can be made as close as we like to L by taking values of x sufficiently close to a (less than a), then we write

$$\lim_{x \to a^-} f(x) = L$$

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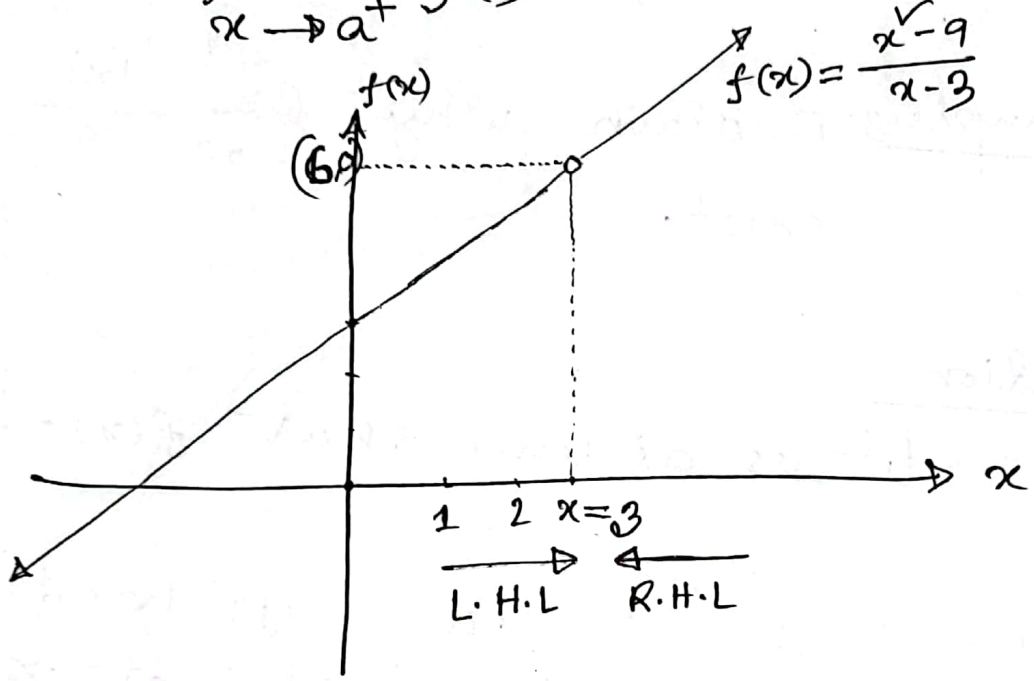
* R.H.L (Right Hand limit)

If the values of $f(x)$ can be made as close as we like to L by taking values of x sufficiently close to a (greater than a), then we write

$$\lim_{x \to a^+} f(x) = L$$



These left hand limit and right hand limit are often called one sided limit.

* Relationship between one-sided and two-sided limits:

The two sided limit of a function $f(x)$ exists at $x=a$ if and only if both of the one-sided limits

exist at $x=a$ and have the same value. i.e,

~~limit~~

$$\lim_{x \to a} f(x) = L \quad \text{if and only if} \quad \lim_{x \to a^-} f(x) = L = \lim_{x \to a^+} f(x)$$

<u>Example 3:</u> Explain why $\lim_{x \to 0} \frac{|x|}{x}$ does not exist.

<u>solution</u>

Let us assume that $f(x) = \frac{|x|}{x}$

Let us find out the left hand limit and right hand limit at $x=0$.

To find out the left hand limit we set,

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} \frac{|x|}{x}$$

$$|x| = \begin{cases} x; & x \geq 0 \\ -x; & x < 0 \end{cases}$$

$$= \lim_{x \to 0^-} \frac{-x}{x}$$

$$= \lim_{x \to 0^-} (-1) = -1$$

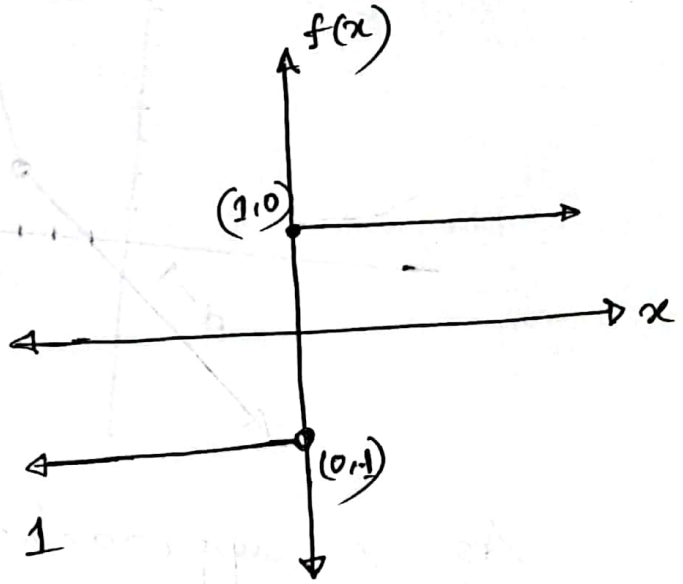
and right hand limit we set,

$$\lim_{x \to 0^+} f(x)$$

$$= \lim_{x \to 0^+} \frac{|x|}{x}$$

$$= \lim_{x \to 0^+} \frac{x}{x}$$

$$= \lim_{x \to 0^+} 1 = 1$$



since, left hand limit and right hand limit are not equal, therefore, $\lim_{x \to 0} \frac{|x|}{x}$ does not exist.

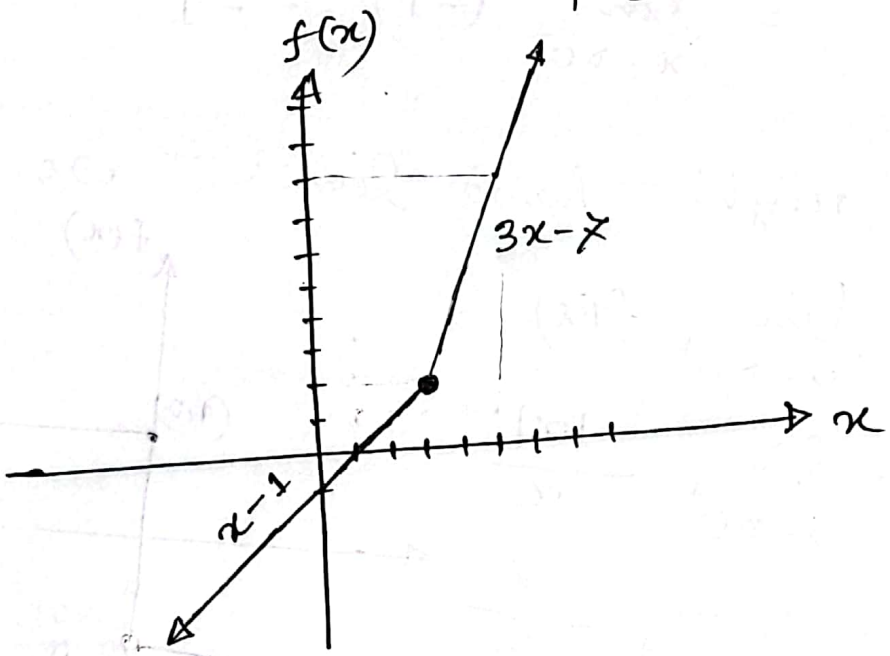
★ Example-4: A function $f(x)$ is defined as

$$f(x) = \begin{cases} x-1 & ; x \leq 3 \\ 3x-7 & ; x > 3 \end{cases}$$

Find $\lim_{x \to 3} f(x)$.

solution: Given that

$$f(x) = \begin{cases} x-1 \; ; & x \le 3 \\ 3x-7 \; ; & x > 3 \end{cases}$$



As x approaches 3 from the left,

$$\lim_{x \to 3^-} f(x) = \lim_{x \to 3^-} (x-1) = 3-1 = 2$$

As x approaches 3 from the right,

$$\lim_{x \to 3^+} f(x) = \lim_{x \to 3^+} (3x-7) = 3 \times 3 - 7 = 2$$

Here we observe that the left hand limit and right hand limit exist and they are equal.

Therefore, $\lim_{x \to 3} f(x) = 2$

* **Example-05:** A function $g(t)$ is defined as

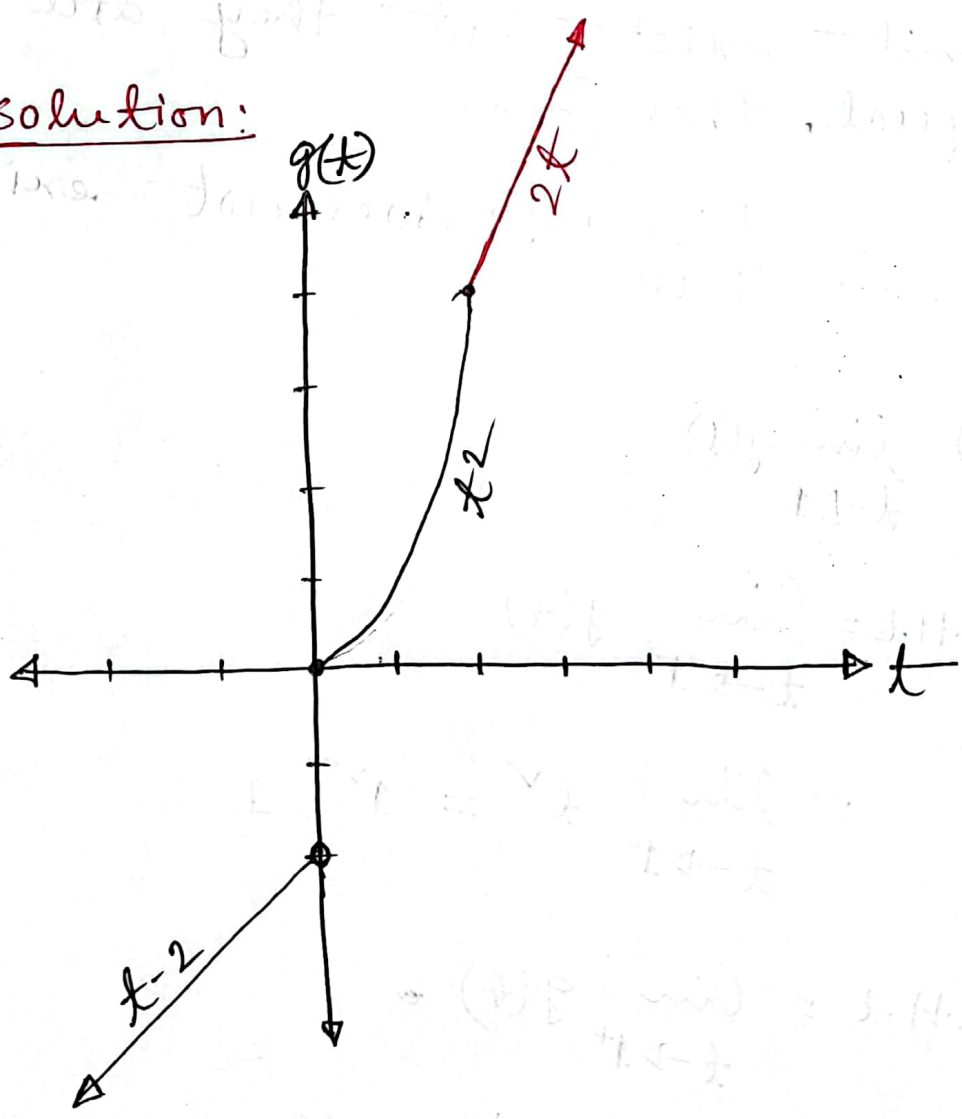
$$g(t) = \begin{cases} t-2 & ; t<0 \\ t^2 & ; 0 \le t \le 2 \\ 2t & ; t>2 \end{cases}$$

Find, a) $\lim_{t \to 0} g(t)$

b) $\lim_{t \to 1} g(t)$

c) $\lim_{t \to 2} g(t)$

Solution:



a) $\lim_{t \to 0} g(t)$

As t approaches 0 from the left,

$$\lim_{t \to 0^-} g(t) = \lim_{t \to 0^-} (t-2) = 0-2 = 2$$

As t approaches 0 from the right,

$$\lim_{t \to 0^+} g(t) = \lim_{t \to 0^+} t^2 = 0^2 = 0$$

The left hand limit and right hand limit exist but they are not equal, therefore,

$\lim_{t \to 0} g(t)$ does not exist.

b) $\lim_{t \to 1} g(t)$

$$L.H.L = \lim_{t \to 1^-} g(t)$$

$$= \lim_{t \to 1^-} t^2 = 1^2 = 1$$

$$R.H.L = \lim_{t \to 1^+} g(t)$$

$$= \lim_{t \to 1^+} t^2 = 1^2 = 1$$

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since, L·H·L = R·H·L implies that limit exists and →

$$\lim_{t \to 1} g(t) = 1$$

c) $\lim_{t \to 2} g(t)$

[L·H·L means Left Hand limit]

$$L \cdot H \cdot L = \lim_{t \to 2^-} g(t)$$

$$= \lim_{t \to 2^-} (t^2) = 2^2 = 4$$

$$R \cdot H \cdot L = \lim_{t \to 2^+} g(t)$$

$$= \lim_{t \to 2^+} (2 \cdot t)$$

$$= 2 \times 2 = 4$$

since L·H·L = R·H·L = 4, so limit exists

and $\lim_{t \to 2} g(t) = 4$

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★ **Homework:** A function $f(x)$ is defined as $f(x) = \begin{cases} \frac{1}{x+2} ; & x < -2 \\ x^2 - 5 ; & -2 < x \leq 3 \\ \sqrt{x+13} ; & x > 3 \end{cases}$

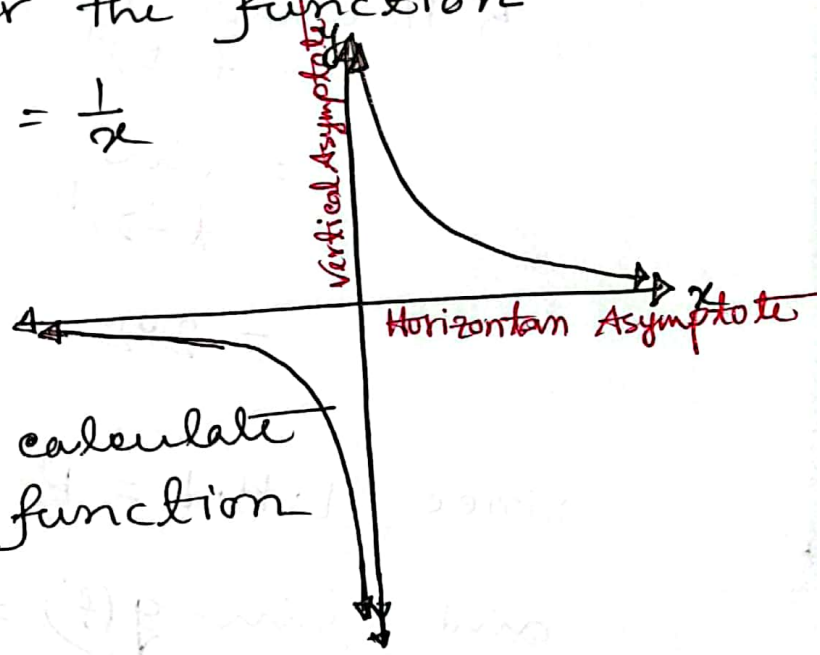
Find, a) $\lim_{x \to -2} f(x)$; b) $\lim_{x \to 0} f(x)$;

c) $\lim_{x \to 3} f(x)$;

★ **Horizontal and Vertical Asymptotes:**

Let us consider the function

$$y = f(x) = \frac{1}{x}$$



Now, we will calculate the limit of function $f(x)$ at $x = 0$

Then,

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} \frac{1}{x} = -\infty$$

$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} \frac{1}{x} = +\infty$$

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In each case, the function $f(x)$ either rises or falls without bound getting closer and closer to the vertical line $x=0$ as x approaches '0' from the sides in the limit. The line $x=0$ is called the vertical asymptote of the curve $y=f(x)=\frac{1}{x}$.

Now let us calculate the limit of the function $f(x)$ when x approaches $+\infty$ and $-\infty$.

Then,

$$\lim_{x \to \infty} f(x) = \lim_{x \to \infty} \frac{1}{x} = \frac{1}{\infty} = 0$$

$$\lim_{x \to -\infty} f(x) = \lim_{x \to -\infty} \frac{1}{x} = \frac{1}{-\infty} = 0$$

Here, $\lim_{x \to +\infty} f(x) = 0$ and $\lim_{x \to -\infty} f(x) = 0$.

Therefore, $y=0$ is the horizontal asymptote of the curve $y=f(x)=\frac{1}{x}$.

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★ Example-06:

Find the horizontal asymptote of $f(x) = \tan^{-1}x$.

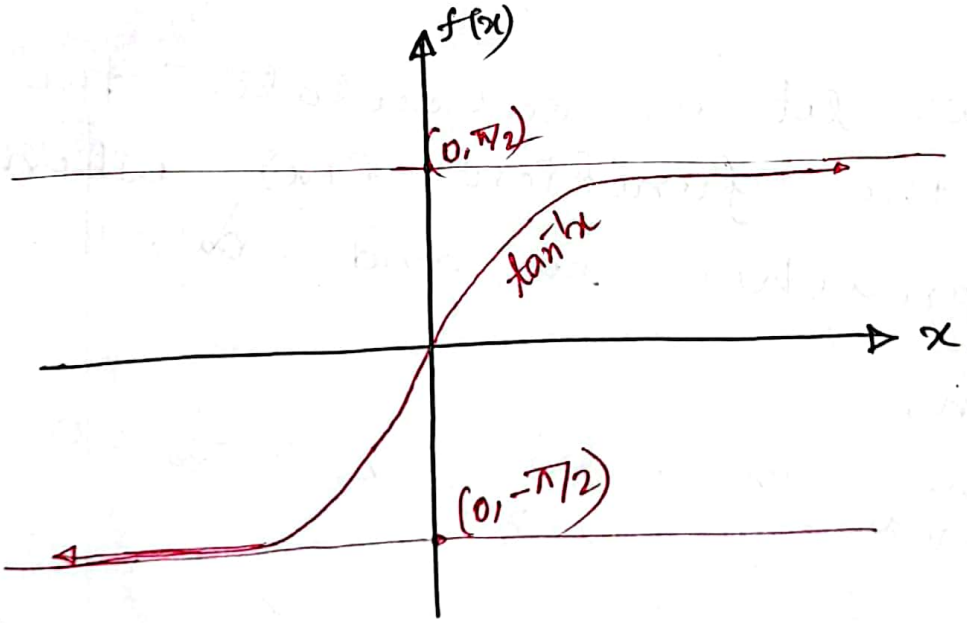
Solution

when x approaches to $+\infty$

$$\lim_{x \to +\infty} f(x) = \lim_{x \to +\infty} \tan^{-1}(x) = \frac{\pi}{2}$$

when x approaches to $-\infty$

$$\lim_{x \to -\infty} f(x) = \lim_{x \to -\infty} \tan^{-1}(x) = -\frac{\pi}{2}$$

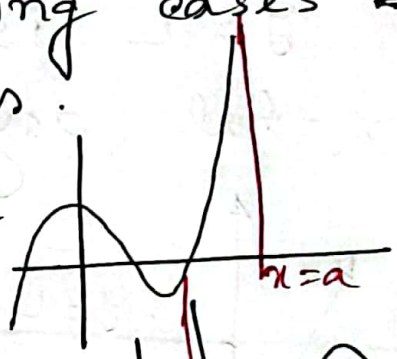


Therefore $y = \pm\frac{\pi}{2}$ are the ~~asymptote~~ horizontal asymptotes of $f(x) = \tan^{-1}x$

* Vertical Asymptote: (see section 1.1 H.A. calculus)

If any of the following cases arrives ~~than~~ such as.

$$\lim_{x \to a^-} f(x) = +\infty \quad , \text{ or}$$



$$\lim_{x \to a^+} f(x) = +\infty \quad , \text{ or}$$

$$\lim_{x \to a^-} f(x) = -\infty \quad , \text{ or}$$

$$\lim_{x \to a^+} f(x) = -\infty$$

then $x = a$ is called Vertical asymptote.

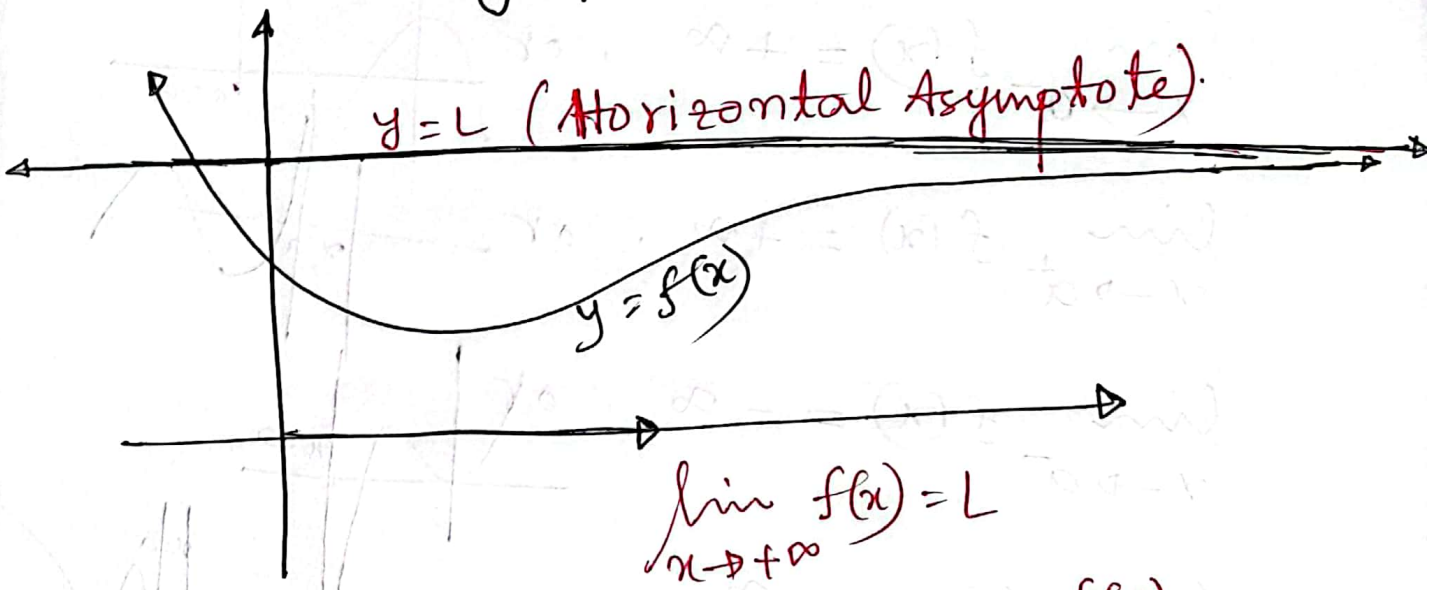
* Horizontal Asymptote:

If the values of $f(x)$ eventually get as close as we like to a number L as x increases without bound, or decreases without bound, then we write,

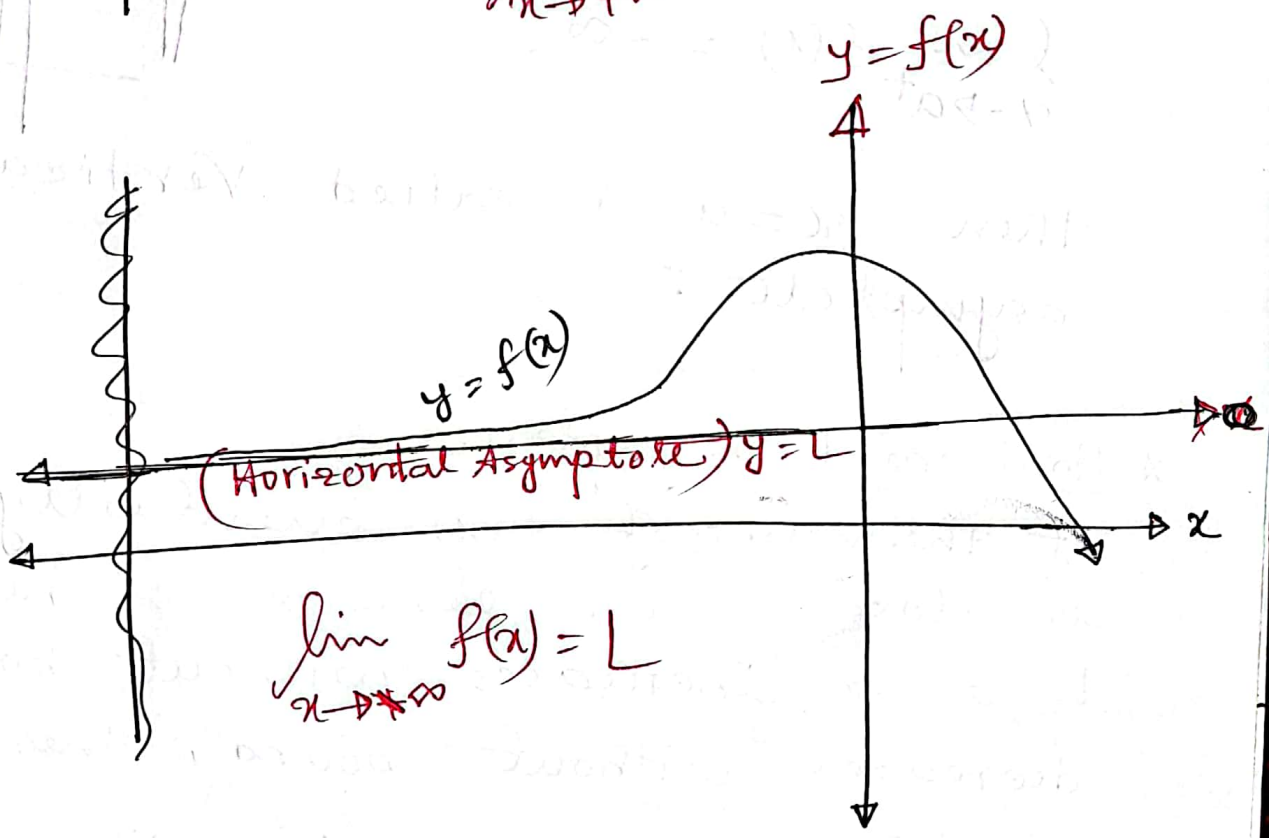
$$\lim_{x \to +\infty} f(x) = L \quad \text{or}$$

$$\lim_{x \to -\infty} f(x) = L$$

In those cases,
$y = L$ is called the horizon-
-tal asymptote of $y = f(x)$



$$\lim_{x \to +\infty} f(x) = L$$



$$\lim_{x \to -\infty} f(x) = L$$

* properties of limit:

Let a be a real number, and suppose that, $\lim_{x \to a} f(x) = L_1$ and $\lim_{x \to a} g(x) = L_2 \neq 0$

Then,

a) $\lim_{x \to a} \left(f(x) \pm g(x) \right) = \lim_{x \to a} f(x) \pm \lim_{x \to a} g(x)$

b) $\lim_{x \to a} f(x)\, g(x) = \lim_{x \to a} f(x) \cdot \lim_{x \to a} g(x)$

c) $\lim_{x \to a} \dfrac{f(x)}{g(x)} = \dfrac{\lim_{x \to a} f(x)}{\lim_{x \to a} g(x)}$

d) $\lim_{x \to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \to a} f(x)} = \sqrt[n]{L_1}$

condition is ~~when~~ $L_1 > 0$.